

Designing a thinking-first AI writing product

Research synthesis, UX recommendation, product plan, and roadmap

Prepared for the Gaddr founding team • April 17, 2026

Core recommendation

Gaddr should be a mode-based writing system, not a single AI canvas. Protect freewriting from interruptions; reveal Gadfly in a structured constellation step; make the guided first-draft coach optional; and end in a genuinely quiet final-draft editor. This approach preserves agency, reduces cognitive overload, and differentiates Gaddr from chat-first writing tools.

Product thesis

Most AI writing tools begin with the machine. Gaddr should begin with the user's own thought, then bring in AI as a challenger, supporter, and researcher. That makes it more valuable for learning, more defensible in education, and more likely to preserve authorship.

Scope	UX best practices, competitor patterns, product design, risks, pricing, roadmap, and experiments.
Bottom line	Ship the freewrite -> constellation -> final draft path first. Treat the first-draft coach as a feature flag until the data proves it is worth the complexity.

Executive summary

The strongest version of Gaddr is not an AI writing assistant that interrupts the user all the time. It is a structured learning workflow that moves through four distinct mental states: generation, sensemaking, drafting, and refinement. Research on interruptions, progressive disclosure, and writing-assistant design all point in that direction [7][8][9][15].

The practical product implication is simple: make the interface change as the user's cognitive job changes. During freewriting, the job is to think and generate. During constellation review, the job is to inspect, compare, challenge, and curate. During drafting, the job is to translate structure into prose. During the final draft, the job is fluency again - but now with higher clarity and stronger evidence.

Recommended product architecture

Four protected modes, with the guided first-draft coach treated as optional rather than structural.

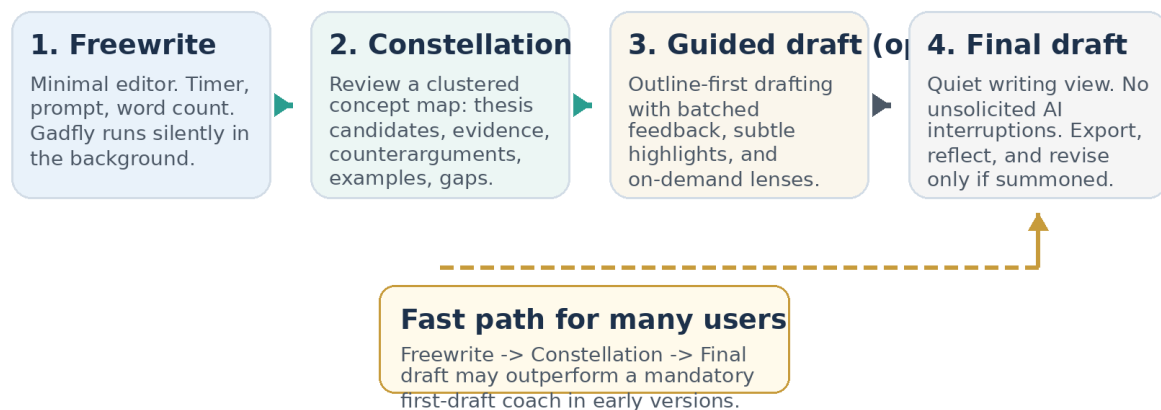


Figure 1. Recommended architecture for Gaddr. Author synthesis based on the cited sources.

What to ship first

- A minimalist timed freewrite editor with optional prompt, adjustable timer, autosave, and no visible AI interruptions.
- A post-timer constellation workspace that clusters Gadfly findings by rhetorical role: claims, evidence, counterarguments, examples, and open questions.
- A pin-and-outline bridge from constellation to writing, so the graph becomes useful material rather than decorative context.
- A truly quiet final-draft editor.
- A guided first-draft coach only as an optional experiment, not as a required middle step.

Decision

For MVP, Gaddr should default to Freewrite -> Constellation -> Final Draft, with optional outlining between the last two steps. The guided first-draft coach should be implemented behind a feature flag and evaluated experimentally before it becomes part of the default flow.

Why this positioning matters

Most adjacent tools either begin with documents and sources (NotebookLM, LiquidText, Heptabase), begin with a spatial canvas (nodepad, Obsidian Canvas, Scrintal), or begin with direct language correction and drafting help. Gaddr's distinctive wedge is different: it begins with unassisted thought, then adds AI only after the user has produced material worth challenging and strengthening [1][2][3][4][5][6].

1. Product thesis and user jobs

Gaddr should be designed as a thinking-first writing system. The user is not asking the AI to produce an essay from nothing. They are using AI to pressure-test, expand, and refine ideas that began as their own.

Primary user jobs

- Help me get past the blank page without being judged too early.
- Show me what I believe, what I am missing, and what the strongest objections are.
- Help me turn scattered thoughts into a defensible argument.
- Support better writing without taking authorship away from me.
- Give me a quiet place to finish once I know what I want to say.

Who this is for first

Best initial audience	Who not to optimize for yet
• College and graduate students writing argumentative essays, reflection papers, and short research-backed assignments. • Ambitious high-school students in AP/IB, debate, model UN, scholarship essays, and admissions writing. • Knowledge workers or creators drafting memos, think pieces, and position papers.	• Mobile-first casual journaling. • Long-form academic citation management with deep library workflows. • Users who only want a one-shot AI writer with no original thought required.

Why the 'learning app' framing is credible

Freewriting research suggests that repeated timed freewriting can improve fluency and confidence. Park found positive effects on confidence and fear of evaluation, though students disliked absent feedback and rigid time limits; Yasuda reported statistically significant gains from repeated 10-minute sessions over a semester [10]. This supports the educational premise of Gaddr - but also suggests the timer should be adjustable and the prompt should remain optional.

2. Research findings that should shape the UX

2.1 Protect focus by separating modes

Nielsen Norman Group's progressive disclosure guidance argues that advanced or infrequently needed features should be deferred to secondary screens, making systems easier to learn and less error-prone [7]. That principle maps almost perfectly onto Gaddr. The freewrite surface should expose only the tools needed for freewriting: timer, prompt, word count, pause/end, and maybe a calm progress cue. Gadfly's complexity should stay hidden until the user has finished the initial thought-generating phase.

This is not just about simplicity. It is about cognitive fit. The writing-assistant design-space literature frames 'writing stage' as a core design dimension and emphasizes agency, ownership, trust, and transparency as central to system design [15]. A single UI that tries to do all of those jobs at once will feel busy, confusing, and overbearing.

2.2 Avoid unsolicited interruption while the user is composing

NN/g warns against workflow-disruptive interfaces, and interruption research consistently finds costs to performance and quality [8][9]. A writing study summary by Foroughi reported that interruptions during writing can reduce essay quality and word count; Katidioti et al. found that self-interruptions slowed completion even more than comparable external interruptions [9].

Implication for Gaddr

Do not 'pester' during freewriting. During guided drafting, never interrupt in the middle of a sentence. Analyze continuously, but reveal feedback only after a pause, paragraph completion, or explicit request.

2.3 Source-grounded AI increases trust

NotebookLM positions itself as a source-grounded research assistant that can answer from the user's sources with inline citations and transform material into study guides, briefings, audio overviews, and mind maps [2]. That is a strong pattern to borrow. Gaddr should never present research findings as anonymous AI wisdom. Every external claim card should disclose where it came from, how recent it is, and whether Gaddr is quoting, summarizing, or inferring.

2.4 Visual maps work best when they support curation, not passive viewing

Concept-map research shows meaningful learning benefits, and the meta-analysis by Schroeder et al. found a moderate positive learning effect overall, with larger gains when learners construct maps rather than merely study them [11]. The implication is important: users should be able to pin, merge, reject, relabel, and connect nodes. A constellation that only looks interesting is weaker than one the user can actively manipulate.

The same research suggests that the map type matters. Mind maps are best for expanding a single topic; concept maps are better for showing complex interrelationships through labeled edges and multiple parents [12]. Because argumentative writing involves support, contradiction, qualification, evidence, and rebuttal, Gaddr should default to a concept-map logic, not a pure mind-map tree.

2.5 Dense graphs are risky

Spatial tools are compelling, but they can also collapse into graph theater. Visualization guidance still favors 'overview first, zoom and filter, then details on demand' [13]. Research on node-link grouping and edge bundling shows that group visualization and bundled edges can reduce clutter, but also that heavy group overlays can hurt accuracy on some tasks [14].

That means Gaddr should not reveal a raw Obsidian-style graph with dozens of moving nodes by default. It should reveal a clustered overview with filters, stable group labels, and a clear details pane.

Recommended constellation view

Use a clustered concept map rather than a dense, force-directed graph. Group by rhetorical role first, then filter by topic or audience lens.

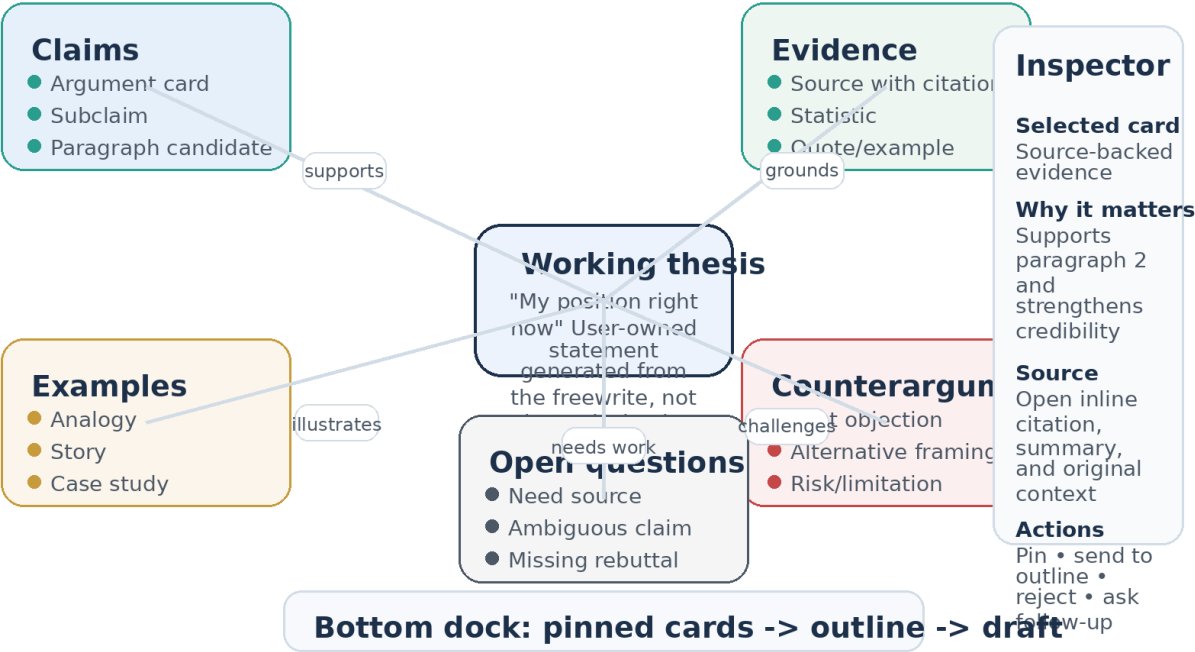


Figure 2. Constellation recommendation: clustered concept map with a source inspector and a pin-to-outline dock.

3. Competitive landscape and what Gaddr should borrow

The adjacent market is crowded, but no major product combines timed freewriting, silent AI research, post-hoc constellation review, optional guided drafting, and a genuinely quiet final-draft mode in one coherent flow. That is Gaddr's opportunity.

Product	What it gets right	Risk to copy blindly	Implication for Gaddr
nodepad [1]	Quiet, background AI that classifies notes, infers links, and surfaces synthesis. Explicit philosophy: augment thinking, not replace it. Desktop-first and spatial by design.	A wide canvas can become too abstract for essay writing. The note model is broader than the needs of a writing-learning workflow.	Borrow the quiet-AI posture and large-screen bias, but add stronger rhetorical scaffolding.
NotebookLM [2]	Source-grounded answers with inline citations; transforms sources into study guides, briefings, and mind maps.	Starts from uploaded materials, not the user's raw thought. More research notebook than writing flow.	Borrow grounded citations, evidence cards, and trust signaling.

Product	What it gets right	Risk to copy blindly	Implication for Gaddr
Obsidian Canvas [3]	Infinite canvas that can mix notes, PDFs, media, and web pages.	Canvas freedom easily becomes clutter and places more burden on the user to impose structure.	Borrow embed flexibility, but constrain the default layout much more.
Heptabase [4]	Strong bridge from whiteboard thinking to article writing; right-sidebar reference workflow is especially relevant.	Heavier knowledge-base orientation. More setup before the user can write.	Borrow the pin-and-write bridge and source sidebar pattern.
Scrintal [5]	Spatial clustering, connected notes, and multi-view note density.	Visual freedom is useful, but can over-index on organization rather than writing throughput.	Borrow clustering and view switching, but keep the editor primary.
LiquidText [6]	Preserves source context while extracting, organizing, and exporting insights.	Document-centric rather than opinion-first.	Borrow 'see context without losing your place' for source inspection.

Competitive positioning

Gaddr should position itself as the anti-chat AI writing tool: write first, think visually second, draft with help third, finish quietly last.

Pricing anchors from adjacent products

The current landscape spans from free core software with optional paid services (Obsidian) to mid-priced research tools (Heptabase at \$8.99/month annual or \$11.99/month monthly) to more expensive bundled AI plans (Google AI Pro at \$19.99/month, with higher NotebookLM access) [2][3][4]. Gaddr can sit comfortably in the \$10-15/month band for individual users if the learning and writing outcomes feel materially better than generic AI chat.

4. Recommended UX architecture for Gaddr

4.1 Mode 1: Freewrite

The first screen should feel almost aggressively minimal. The purpose is to generate original language before the system starts steering. Users can start from a blank page or an optional 'prompt of the day.' They choose a timer - recommended presets: 5, 7, and 10 minutes, with 7 as the default. The countdown should be quiet: visible but not dominant, and optionally hidden until the last 30 seconds for anxious users [10].

- Visible controls: prompt, timer, pause, finish early, word count, and maybe a calm autosave state.
- Not visible during the timer: grammar flags, counterarguments, source cards, node graphs, AI rewrites, or pop-up suggestions.

- Background behavior: Gadfly can chunk the text during idle moments, extract propositions, identify stance and uncertainty, and precompute evidence/counterargument cards.
- Safety rule: no AI-generated prose is inserted into the writing area during freewrite.

4.2 Transition: from freewrite to constellation

The timer should not end with a jarring hard cut. A better transition is: time ends -> freeze the current text -> show a one-screen summary with thesis candidates or 'what Gadfly noticed' -> then reveal the constellation. This preserves a sense of continuity and makes the reveal feel earned rather than abrupt.

Suggested microcopy at transition

"You wrote the raw material. Gadfly looked for tensions, evidence, objections, and next questions. Review the constellation before you draft."

4.3 Mode 2: Constellation

This should be the signature experience. But it must be more useful than a pretty graph. The recommended default is a clustered concept map with the user's working thesis in the center, rhetorical-role clusters around it, a right-side inspector for source context, and a bottom dock for pinned cards that become the outline.

Drafting workspace recommendation

Make the editor primary. Keep Gadfly in the side rail and bottom tray so the user's text remains the visual center of gravity.



Figure 3. Editor-forward workspace for the guided-draft mode.

- Primary clusters: Claims, Evidence, Counterarguments, Examples, Open Questions.
- Secondary filters: topic, source type, audience lens, confidence, and recency.
- Card actions: pin, reject, merge duplicate, ask follow-up, add note, send to outline.
- Evidence cards should show source title, publication, date, and confidence / relevance cues.
- Graph motion should be limited. Auto-layout once on reveal, then stay stable unless the user explicitly re-groups.

4.4 Mode 3: Guided drafting (optional)

This is the most debatable step in your current concept, and I would not make it mandatory. The research does support targeted revision feedback, especially when it is detailed and sentence-level rather than absent or purely binary [17]. But that does not mean the right move is to bombard the user while they are trying to produce a first draft.

The best compromise is to make guided drafting available as a mode, not a requirement. In this mode, the writing area stays primary and Gadfly feedback appears in a side rail or as gentle, categorized highlights after a pause. Categories should be rhetorical rather than generic: evidence gap, strongest objection missing, logic leap, overclaim, unclear referent, or tone mismatch [15][17].

- No pop-ups while the cursor is moving.
- Feedback batched after paragraph completion or explicit refresh.
- Inline highlighting should be subtle and ignorable.
- One-click 'show me the skeptical reader view' or 'show me the strongest support' lenses can build on the persona literature [18].

4.5 Mode 4: Final draft

This mode should feel almost sacred. Once the user enters final draft, unsolicited AI should go quiet. The user can still summon Gadfly manually, open pinned cards, or inspect sources, but the default should be uninterrupted writing. This is the moment when many writing tools fail: they keep acting like the brainstorming stage never ended.

5. Constellation design options

You specifically asked what the optimal design for the node/constellation step might be. There is no universal best choice, but some options are much better fits for argumentative writing than others.

Design option	Strengths	Weaknesses	Fit for Gaddr
Pure force-directed graph	Feels alive, exploratory, and visually differentiated.	Usually becomes cluttered fast. Relationship semantics are weak. Hard to move from insight to prose.	Low
Mind map tree	Simple and easy to parse.	Too single-threaded for evidence, counterarguments, and cross-links. Edges are usually unlabeled [12].	Medium for brainstorming; weak for argumentation
Clustered concept map	Supports multiple parents, labeled relationships, and rhetorical structure. Better fit for argumentation [11][12].	Needs careful density management and strong defaults.	High
Card grid + minimap	Fast to scan, easier on smaller screens, less graph clutter.	Can lose the feeling of relationships across cards.	Medium-high

Design option	Strengths	Weaknesses	Fit for Gaddr
Split-screen editor + side rail + optional map	Best for throughput. Keeps writing primary and lets the map serve the draft.	Less visually magical than a full-canvas experience.	Very high for production use

Recommendation

Use a clustered concept map as the sensemaking surface, but pair it with a fixed inspector and a pin-to-outline dock. That makes the map useful enough for thinking and concrete enough for writing.

Recommended clustering logic

- Primary grouping: rhetorical role. This is the most useful default for essay writing.
- Secondary grouping: theme / subtopic, produced by semantic clustering.
- Optional tertiary filter: audience lens (teacher, skeptic, policymaker, supporter) or source type (user thought, Gadfly inference, external evidence).
- Rank within cluster by relevance to the current working thesis, not by graph centrality alone.

Suggested node types

- User thought
- Possible thesis
- Claim / subclaim
- Evidence / source-backed fact
- Counterargument
- Rebuttal
- Example / analogy
- Open question / gap
- Paragraph candidate

Suggested edge labels

Do not draw generic lines. Use support relationships with labels: supports, challenges, qualifies, defines, illustrates, contradicts, requires evidence, and rebuts. This is where concept-map logic pays off [12].

6. Product rules, interaction details, and pain points to avoid

Non-negotiable UX rules

1. The user's text remains the center of gravity in every mode except full-screen constellation review.
2. AI suggestions are clearly differentiated from user-authored text [15].
3. External claims are source-grounded or explicitly marked as inference [2][15].
4. Mode changes are explicit. Users always know whether they are freewriting, reviewing, drafting, or finishing.
5. Nothing is auto-accepted into the essay. The user must actively pin, insert, or apply it.

Pain points you are likely to hit

Likely failure modes	Design responses
• Graph theater: beautiful network, low drafting value. • Premature AI takeover: the assistant starts writing before the user knows what they think. • Notification fatigue: too many nudges erode trust and flow. • Hallucinated research or unattributed claims. • Too much mode-switching on small screens.	• Tie the graph to pinning, outlining, and paragraph building. • No AI prose insertion during freewrite; no default ghostwriting. • Batch suggestions and let users choose when to review them. • Require source attribution for external evidence and clearly mark inferences. • Build desktop-first. Treat mobile as a capture companion, not the full experience.

What to do about the user's original freewrite

Do not hide it behind the graph. The freewrite should remain available in a collapsible panel or as linked source chunks inside the constellation. Users need to see where their own thinking started, both for identity and for coherence. One especially strong pattern would be to segment the freewrite into proposition cards that each link back to their original passage. That turns the freewrite into a first-class artifact rather than disposable scratch text.

Recommended AI behaviors by mode

Mode	Allowed background help	Should stay off by default
Freewrite	Silent proposition extraction, topic modeling, stance detection, evidence search, counterargument search	Suggestions, rewrites, grammar flags, source popups, node animation
Constellation	Summaries, source cards, cluster labels, thesis candidates, ask-follow-up interactions	Auto-writing paragraphs into the user's draft
Guided draft	Batched rhetorical feedback, source recall, on-demand lenses, outline-to-paragraph support	Interruptive popups or constant red underlines
Final draft	Manual summon only	Unsolicited assistance by default

A note on personas / lenses

The writer-defined AI personas work is especially relevant if you want Gadfly to challenge, support, or interpret the draft from different perspectives. The study found that writers valued on-demand personas for getting different perspectives, although feedback could become verbose and hard to define initially [18]. For Gaddr, that suggests a curated lens library rather than a blank persona builder in the first release.

- Challenge me
- Support me
- Skeptical teacher

- Friendly reader
- Opposing columnist
- Policy reviewer

7. Business model and go-to-market

Positioning

Gaddr should be positioned as a premium learning and writing product, not as a generic AI chatbot. The message is not 'write with AI.' It is 'think first, then write better with AI.' That angle is particularly strong in education, where authorship and academic integrity matter.

Initial wedge

- Essay-heavy students who need to turn opinion into argument.
- Writing-center and debate-club pilots that can validate learning outcomes.
- Creators and knowledge workers who publish argument-driven writing and value thought ownership.

Recommended pricing

A sensible launch structure is freemium + Pro + educator / team plans. This sits naturally against adjacent tools: Heptabase in the high-single-digit to low-teens range, Obsidian's free core with optional services, and more expensive AI bundles such as Google AI Pro [2][3][4].

Tier	Price	What it includes	Why it exists
Free	0	A limited number of sessions per month, basic prompt library, basic constellation, export watermark or limited history	Acquire users and demonstrate the core aha moment
Pro	12/month annual or 15/month monthly	Unlimited sessions, richer Gadfly research, source grounding, guided draft mode, export and version history	Main individual revenue tier
Edu / Team	8-10/user/month or pilot pricing	Shared prompts, instructor libraries, assignment templates, basic analytics, privacy controls	Institutional expansion after product fit
Power / Studio (optional later)	20-25/month	Higher research limits, advanced lenses, deep export, collaboration	Upsell for creators and power users

Go-to-market

- Lead with a simple thesis: 'Write before AI.' That is memorable and distinct.
- Show real before-and-after essays or position papers created through the flow.
- Pilot with writing centers, AP Lang / rhetoric teachers, debate coaches, and fellowship / scholarship essay communities.
- Use social proof around thought ownership, not around 'the AI wrote this for me.'

Metrics that matter

- Freewrite completion rate
- Constellation reveal rate
- Pin-to-outline or pin-to-draft rate
- Final draft completion rate
- 7-day and 30-day retained writers
- Source inspection rate (proxy for trust and rigor)
- Improvement in rubric scores or self-reported clarity in pilots

8. Roadmap and experimentation plan

Phase 0 - Prototype (0 to 6 weeks)

- Desktop web prototype.
- Timed freewrite with autosave.
- Background Gadfly pipeline that extracts claims, questions, and evidence gaps.
- Static post-write constellation mock with source-backed cards.
- Five-user usability tests focused on mode transitions.

Phase 1 - MVP (6 to 16 weeks)

- Working constellation with cluster filters and source inspector.
- Pin-to-outline and export.
- Final-draft quiet editor.
- Trust features: citations, recency, source metadata, and explicit inference labels.
- Basic analytics for completion, pinning, and retention.

Phase 2 - Product v1 (4 to 8 months)

- Optional guided-draft mode with categorized rhetorical feedback.
- Lens library (challenge, support, skeptical teacher, policymaker, etc.).
- Paragraph candidate cards and outline-to-draft bridging.
- Reflection / compare view between freewrite, first draft, and final draft.
- Improved onboarding based on usability findings.

Phase 3 - Education and expansion (8 to 12 months)

- Instructor prompt packs and assignment templates.
- Shared team / class spaces.
- Privacy and academic-integrity controls.
- Outcome studies with partner classes or writing centers.
- Mobile capture companion for collecting thoughts, not for full constellation use.

Experiments worth running early

Experiment	Primary metric	Hypothesis
Mandatory guided draft vs optional guided draft	Outline use, draft completion, session satisfaction	The guided coach helps, but only if it does not become friction.

Experiment	Primary metric	Hypothesis
5 vs 7 vs 10 minute defaults	Completion rate, word count, self-reported stress	7 minutes may be the best compromise between momentum and pressure [10].
Clustered map vs card grid default	Pin rate, time to first useful action, confusion	Pretty graphs may underperform simpler structured views.
Live highlights vs batched feedback	Draft throughput, quality ratings, frustration	Batched feedback should preserve flow better [8][9][17].
Blank prompt vs prompt of the day prominence	Start rate, freewrite completion, retention	Prompts help some users, but should never be required [10].

Architectural principle

Treat every post-constellation surface as modular. If the guided-draft coach underperforms, you should be able to remove it cleanly and route users directly into the final-draft editor without rethinking the rest of the product.

9. Risks, open questions, and final recommendation

Open questions

- How much of the user's freewrite should be segmented automatically versus left intact?
- Will users prefer a full constellation canvas or a simpler cards-first review screen in practice?
- Does guided drafting improve final essays enough to justify the complexity cost?
- How much external research should Gadfly fetch automatically before trust or latency become problems?
- What level of source grounding is necessary for classroom acceptance?

What I would do if I were building Gaddr now

- Build desktop-first.
- Make freewrite beautifully simple.
- Invest heavily in the constellation reveal and the pin-to-outline bridge.
- Launch with a quiet final-draft editor.
- Keep guided drafting behind a flag until user data proves it improves outcomes.
- Position the whole product around ownership, rigor, and learning rather than convenience alone.

Final recommendation

The best version of Gaddr is a deliberately staged product: unassisted thought first, AI-augmented sensemaking second, optional guided drafting third, uninterrupted finishing last. If you nail that sequence - especially the transition from freewrite to constellation and from constellation to prose - you will have a product that feels genuinely different from both chatbots and note-taking canvases.

Selected sources

The report above synthesizes current product pages, pricing/help pages, UX articles, and research papers that were directly relevant to Gaddr's flow. Accessed April 17, 2026.

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Method note

I prioritized official product pages and primary sources where possible, then used HCI / UX research to translate those patterns into design recommendations for Gaddr. Product recommendations in this report are my synthesis, not direct claims from any single source.